



Policy-sensitive crypto assets: Event study of thematic returns around U.S. elections[☆]

Kevin Pirazzi Maffiola^{a,*}, Hasret Ozan Sevim^{a,b}

^a Catholic University of the Sacred Heart, School of Banking, Finance and Insurance Sciences, Largo A. Gemelli 1, 20123 - Milan, Italy

^b International School of Advanced Studies, University of Camerino, c/o Polo delle Scienze, Via Gentile III Da Varano, 62032 Camerino, Italy

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ABSTRACT

This paper analyzes how thematic crypto asset categories responded to the 2024 U.S. presidential election using cumulative abnormal returns (CARs) within an event study framework. We examine four thematic categories: Made in U.S., DeFi & Real-World Assets, AI & Big Data, and World Liberty Financial. The results show significant negative CARs in the pre-election window, followed by sharp positive reversals on the election day and after the election. The most pronounced post-election recoveries appear in the Made in U.S. category with CARs over 40 %, suggesting market reassessment of regulatory expectations. Innovation-driven categories such as (World Liberty Financial, DeFi & Real-World Assets, and AI & Big Data) exhibit more muted responses. These findings underscore how political alignment influences crypto performance and offer new evidence on the pricing of political risk in digital asset markets.

1. Introduction

The crypto asset market has emerged as a dynamic and rapidly evolving segment of global finance, marked by high volatility, decentralized governance, and pronounced sensitivity to external shocks. Unlike traditional financial assets, crypto assets often react sharply to regulatory developments, political uncertainty, and macroeconomic events. Prior studies have documented significant market responses to regulatory announcements (Borri and Shakhnov 2020; Shanaev et al., 2020; Yen et al., 2023), cybersecurity incidents (Chen et al., 2023), institutional product launches (Babalos et al., 2025; Mazur and Polyzos, 2024), and macroeconomic shocks (Corbet et al., 2020). Despite exhibiting semi-strong form efficiency, crypto asset markets remain segmented, with divergent responses across different asset categories (Myeong and Kim, 2025).

Recent regulatory interventions, coupled with the approval of spot Bitcoin and Ethereum ETFs, have further integrated digital assets with traditional finance, intensifying their sensitivity to political developments (Chen et al., 2025; Marshall and Nae, 2025). However, while the effects of political elections on traditional markets are well studied, their impact on heterogeneous segments within the crypto ecosystem remains underexplored.

This study addresses that gap by analyzing how thematic categories of crypto assets responded to the 2024 U.S. presidential election¹, a watershed event for digital asset regulation and institutional adoption (Rogers et al., 2024). Using an event study methodology, we assess cumulative abnormal returns (CARs) for four categories: (i) Made in U.S., (ii) World Liberty Financial (WLF), (iii) DeFi & Real-World Assets (RWA), and (iv) AI & Big Data. These categories capture key structural and regulatory dimensions of the crypto landscape (policy and

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* Corresponding author.

E-mail addresses: kevin.pirazzimaffiola@unicatt.it (K. Pirazzi Maffiola), hasretozan.sevim@unicatt.it (H.O. Sevim).

¹ The election was on November 5th, 2024, and the results were available as of November 6th. Link to the article published by Bennet B. on the Time on November 6th, 2024: <https://time.com/7173338/donald-trump-wins-election-2024/>

innovation), advancing beyond market-based classifications (Babalos et al., 2025; Saggu et al., 2025; Zhou, 2025).

Our findings reveal three key dynamics: (1) anticipatory declines in the pre-election window, consistent with political uncertainty (Miba'am and Güngör, 2025); (2) sharp positive reversals post-election (election day and long term) as markets reassessed regulatory outlooks; and (3) persistent heterogeneity across categories, with the policy-sensitive category Made in U.S. exhibiting the strongest and most sustained CARs.

These results contribute novel empirical evidence on how political signals are priced in crypto asset markets, highlight the role of thematic positioning in managing electoral risk, and provide useful insights for investors and policymakers navigating crypto's increasing financial integration.

The remainder of the paper is organized as follows. Section 2 details the sample and methodology, followed by the results section. Section 4 presents the discussion and conclusion.

2. Data and methodology

2.1. Sample

Prior crypto assets event studies have predominantly analyzed Bitcoin or relied on restrictive selection criteria, including exchange-specific listings (Babalos et al., 2025), market-cap based (Zhou, 2025), or regulatory classifications (Saggu et al., 2025). We collected daily price data from CoinGecko's API (Saggu et al., 2025). Bitcoin serves as our market benchmark following standard practice (Meyer and Ante, 2020), while being excluded from thematic categories to avoid overlap. We advance this literature by constructing a novel dataset of tokens across four thematic categories (see Appendix 1) that capture key developments in the crypto ecosystem.

This classification system provides three key advantages: (1) it reflects structural shifts in crypto markets beyond simple market-cap rankings, (2) enables analysis of regulatory sensitivity across policy-relevant and innovation categories, and (3) captures their growing thematic diversification.

2.2. Event study methodology

This study employs an event study framework to investigate market reactions of different thematic categories of crypto assets to the 2024 U.S. presidential election. The methodology follows established practices in financial economics (MacKinlay, 1997; Boehmer, 1991). The analysis, which is three-staged and consistent across all specific categories and the Overall sample (all studied crypto assets), is as follows: (1) expected returns specification, (2) model parameters estimation, and (3) calculation and aggregation of abnormal returns.

The foundation of our analysis is the market model specified in Eq. (1), which estimates expected returns by accounting for systematic market movements.

$$E(R_{i,t}) = \hat{\alpha}_i + \hat{\beta}_i R_{m,t} \quad (1)$$

where:

- $E(R_{i,t})$ represents the expected log daily return of crypto asset i at time t
- $R_{m,t}$ denotes Bitcoin's log daily return.
- $\hat{\alpha}_i$ is the estimated alpha for asset i and represents the asset's excess return not explained by the market's movement.

Table 1

Descriptive statistics.

Category	Mean	Standard Deviation	Max.	Min.
Overall	0.33 %	4.43 %	8.76 %	−9.02 %
Made in U.S.	0.46 %	4.31 %	9.52 %	−9.17 %
WLF Portfolio	0.26 %	4.47 %	9.74 %	−9.98 %
DeFi & RWA	0.31 %	4.80 %	9.62 %	−9.72 %
AI & Big Data	0.25 %	4.91 %	10.20 %	−9.98 %

- $\hat{\beta}_i$ is the estimated beta for asset i and measures the sensitivity of crypto assets i to Bitcoin.

The use of Bitcoin as the market proxy is justified by three key factors, allowing us to isolate category-specific effects from general market trends.: (1) its status as the first and dominant crypto asset, (2) its substantial market capitalization relative to other crypto assets, and (3) its well-documented correlation with the broader crypto market (Meyer and Ante, 2020; Saggu et al., 2025).

We estimate model parameters ($\hat{\alpha}_i$ and $\hat{\beta}_i$) for each crypto asset using ordinary least squares (OLS) regression over a 140-day estimation window spanning from day -150 to day -10 relative to the event date (November 6, 2024), aligned to the approach of Saggu et al. (2025). First, this extended period serves several purposes; it offers enough data for stable and reliable estimates, addressing concerns about estimation sensitivity (Armitage, 1995). Second, the 10-day buffer between the estimation window and event date prevents contamination of our parameters from anticipatory market movements related to the election, and third, the window length balances the need for estimation precision with stable market conditions.

The calculation of abnormal returns $AR_{i,t}$ for each crypto asset i on day t is computed with the model in Eq. (2):

$$AR_{i,t} = R_{i,t} - E(R_{i,t}) \quad (2)$$

$AR_{i,t}$ represent the portion of an asset's return that cannot be explained by general market movements, thus isolating the specific impact of the election event on each crypto asset.

AAR_t as shown in Eq. (3), represents the average abnormal returns for a specific day and is calculated by adding the $AR_{i,t}$ for a day t divided by the number of crypto assets N .

$$AAR_t = \frac{1}{N} \sum_{i=1}^N AR_{i,t} \quad (3)$$

To measure the total effect of the event over the specified period, we calculated Cumulative Average Abnormal Returns (CAARs), which are obtained by summing the daily Average Abnormal Returns (AAR_t) within the event window. For this study, we refer to this metric as Cumulative Abnormal Returns (CARs).

We examine CARs for:

- Pre-event windows: $[-10, -1]$ days to capture anticipatory effects,
- Pre- and post-event window $[-1, 1]$ and $[-1, 0]$ days to capture the effect around the event.
- Immediate post-event windows: $[0, 2]$ and $[0, 6]$ days to assess initial market reactions,
- Extended post-event windows: $[0, 30]$ days to evaluate longer-term impacts.

This comprehensive approach allows us to document the complete evolution of market reactions, from initial uncertainty through

Table 2
Results based on winsorized sample at the 5th and 95th percentile levels.

	Overall			Made in U.S.			WLF Portfolio			DeFi & RWA			AI & Big Data		
	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.
Event windows before the event (excluding the event day)															
Pre- and post-event window															
CAR [-10;-1]	-4.66 %	-6.65	***	-4.79 %	-3.27	***	-1.77 %	-1.12	***	-4.93 %	-5.52	***	-3.86 %	-1.81	*
CAR [-1;1]	7.44 %	13.60	***	6.57 %	10.52	***	9.39 %	8.53	***	8.26 %	8.29	***	6.82 %	5.82	***
CAR [-1;0]	14.92 %	28.58	***	14.14 %	16.07	***	16.35 %	13.10	***	15.47 %	21.36	***	15.71 %	13.60	***
Event windows after the event (including the event day)															
CAR [0;2]	1.78 %	2.99	***	1.26 %	1.14	***	2.80 %	1.47	***	3.48 %	3.79	***	-1.44 %	-1.59	***
CAR [0;6]	2.02 %	1.97	***	7.21 %	3.36	***	3.66 %	2.56	***	-1.76 %	-1.56	***	-0.98 %	-0.49	***
CAR [0;30]	32.52 %	10.02	***	40.85 %	7.10	***	24.02 %	3.00	***	30.81 %	5.43	***	30.80 %	5.45	***
Event windows after the event (excluding the event day)															
CAR [1;2]	-6.87 %	-11.84	***	-6.33 %	-5.20	***	-6.94 %	-3.51	***	-6.14 %	-7.02	***	-9.61 %	-12.98	***
CAR [1;6]	-6.64 %	-5.92	***	-0.37 %	-0.16	***	-6.08 %	-4.54	***	-11.37 %	-9.45	***	-9.15 %	-4.32	***
CAR [1;30]	23.86 %	7.11	***	33.26 %	5.41	***	14.29 %	1.85	*	21.19 %	3.68	***	22.62 %	-8.50	***

Notes: *** denotes a statistical significance at a p-value < 0.01, while ** and * denote a statistical significance at a p-value of respectively < 0.05 and < 0.10. The daily ARs have been winsorized at the 5th and 95th percentiles.

subsequent price adjustments as regulatory expectations become clearer.

The descriptive statistic for daily ARR_t is presented in Table 1.

3. Results

The results of *t*-tests across various event windows are presented in Table 2 and is based on a sample in which daily abnormal returns have been winsorized at the 5th and 95th percentiles. Overall, we find strong evidence of anticipatory market behavior, with statistically significant negative CARs of -4.66 % (p-value < 0.01) in the pre-election window [-10,-1]. This decline reverses sharply around the event, with strong positive CARs of +14.92 % (p-value < 0.01) emerging in the [-1,0] window. Immediate positive CARs of +1.78 % (p-value < 0.05) are sustained in the [0,2] period and in the subsequent [0,6] period. The positive trend continues to grow, leading to sustained gains of +35.52 % (p-value < 0.01) over the longer post-election period [0,30]. The results show that excluding the event day from the post-event window estimation in the first few subsequent days, CAR declines to then rise in the longer window.

Cross-category heterogeneity is notable. The Made in U.S. category exhibits the strongest dynamics, with pre-election CARs of -4.79 % (p-value < 0.01) followed by +40.85 % (p-value < 0.01) in the event window [0,30], suggesting acute sensitivity to policy shifts.

Among innovation-focused categories, DeFi & RWA assets exhibit the most negative pre-election CARs (-4.93 %, p-value < 0.01) but CARs rebound strongly in the [0,30] window (+30.81 %, p-value < 0.01). AI & Big Data show a negative pre-event CARs (-3.80 %, p-value < 0.10) and aligned post-election (+30.80 %, p-value < 0.01). The WLF category shows the lowest decline pre-event at -1.77 % and the lowest post-event gain at +24.02 % in the window [0,30].

Appendix 2 presents intra-day event windows. Generally, most positive CARs around the event are in the 24 h preceding it, with a significant portion of the contribution of the final hour. In contrast, a smaller proportion of positive returns occur in the 24 h after the event, primarily concentrated before the close of the event day.

We conducted two additional analyses to test the robustness of our findings. First, we ran the model on a sample winsorized at the 1st and 99th percentiles and on a non-winsorized sample. The results, which largely confirm our baseline model's findings, are presented in Appendices 3 and 4. Secondly, a mean-adjusted model was employed to re-estimate abnormal returns, confirming our main results. In this specification, the expected returns $E(R_i, t)$ were established using a constant mean return model calculated as the arithmetic mean of the asset's daily log returns over the 140-day estimation window ($t=-150$ to $t=-10$). Finally, we performed a confirming analysis (available upon request), excluding assets that are subject to a secondary classification among our categories, which is aligned to baseline methodology.

4. Conclusion and discussion

Our analysis reveals an anticipation-correction pattern in crypto markets around the 2024 U.S. election. We found significant pre-election declines followed by sharp post-election rebounds on Election Day and over the subsequent 30 days. This pattern demonstrates the market's rapid processing of political uncertainty and policy signals (Miba'am and Güngör, 2025).

Moreover, we identify substantial heterogeneity across thematic categories. Policy-sensitive assets, particularly the Made in U.S. category, showed the strongest CARs, suggesting investors reward perceived regulatory alignment. In contrast, innovation-driven categories (DeFi & Real-World Assets, AI & Big Data, and World Liberty Financial) exhibited more muted post-election responses, indicating less sensitivity to U.S.-specific political developments.

For policymakers, our results show that crypto markets rapidly incorporate regulatory expectations, with price movements acting as

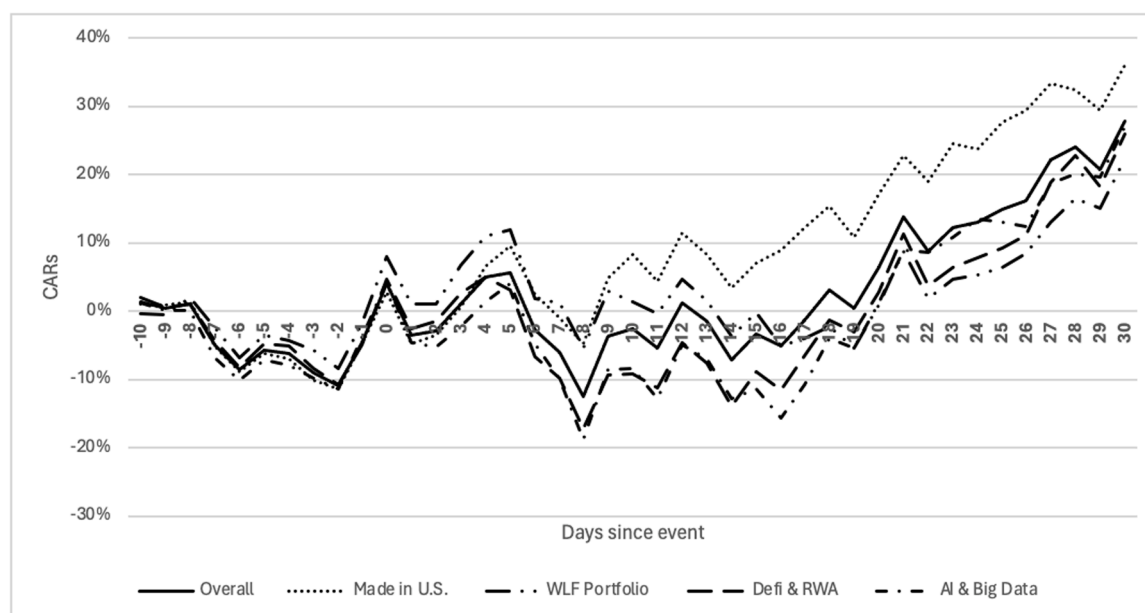


Fig. 1. CARs over the event window.

real-time indicators of policy impact. The outperformance of Made in U.S. crypto assets suggests markets reward regulatory alignment, with implications for designing electoral communication strategies to mitigate volatility.

Our findings highlight opportunities for investors to generate alpha through thematic positioning around political events. While our analysis of a U.S. election provides clear insights, the generalizability to other jurisdictions and long-term effects requires further research. Cross-country comparisons and extended event windows are key avenues for future work.

Further research should investigate crypto market responses to political shocks and the mechanisms through which policy signals, like the announcement of a U.S. strategic crypto reserve² drive crypto returns. Our findings show that crypto markets now process political information with semi-strong efficiency, but retain distinct pricing dynamics across different segments, indicating growing integration into traditional financial markets.(Fig 1)

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Data availability

The data used in this study originate from a third-party provider and are not supplied by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1

Table A2

Table A3

Table A4

Table A5

² Link to website of the White House related to the strategic crypto reserve: <<https://www.whitehouse.gov/presidential-actions/2025/03/establishment-of-the-strategic-bitcoin-reserve-and-united-states-digital-asset-stockpile/>>

Table A1
Crypto Assets Thematic Categories Composition.

Category	Crypto Asset Tickers	Description
DeFi & RWA	LINK, OM, UNI, CFG, RIO, RPL, ANKR, MPL, ENA, MKR, JUP, CRV, XVS, JOE, QI, LDO, CAKE, PENDLE, JTO, RAY, COMP, BAL, GNO, CVX, SNX, ZRX, YFI, SUSHI, ETHFI	This category specifically focuses on well-established DeFi tokens and newer projects linked to DeFi, which have witnessed growth due to innovations in decentralized lending, yield farming, and governance mechanisms. The initial dataset was constructed by identifying all tokens listed on at least one of the five major exchanges at the time of the event. From this universe, tokens were assigned to the DeFi & RWA category based on their market tickers and exchange-provided sector labels. This preliminary classification was then verified by consulting each project's whitepaper to confirm that its core utility and function aligned with decentralized finance or real-world asset tokenization. This two-stage process ensured an accurate and substantiated final sample.
AI & Big Data	TAO, RENDER, FIL, FET, GRT, AKT, GLM, LPT, ARKM, IO, FLUX, RLC	These assets are associated with the fields of artificial intelligence and big data. Multiple of these projects are increasingly being integrated with blockchain-based solutions for decentralized AI training and data processing applications. The initial dataset was constructed by identifying all tokens listed on at least one of the five major exchanges at the time of the event. From this universe, tokens were assigned to the AI & BigData category based on their market tickers and exchange-provided sector labels. This preliminary classification was then verified by consulting each project's whitepaper to confirm that its core utility and function aligned with decentralized finance or real-world asset tokenization. This two-stage process ensured an accurate and substantiated final sample.
Made In U.S.	XRP, SOL, DOGE, ADA, AVAX, HBAR, ARB, XLM, BCH, LTC, APT, NEAR, KAS, ALGO, OP, EOS, STX, BONK, WLD, INJ, SEI, THETA	Tokens were considered to have a U.S. nexus if their whitepapers or websites pointed to a U.S. headquarters, or if the founders and core developers were primarily based in the country. This method recognizes that even with a decentralized network, the primary control or development team are related to U.S.-based entities, thereby establishing a clear jurisdictional link.
WLF Portfolio	ETH, TRX, LINK, SUI, ONDO, AAVE, ENA, STETH, WBTC	This category represents the portfolio (at the time of the event) of the World Liberty Financial^a, a cryptocurrency venture launched in September 2024 by the current U.S. president and can be monitored by Arkham Intelligence^b.

Notes: This table includes crypto assets that were actively traded on at least one of five major exchanges (Binance, Coinbase, Kraken, OKX, and Bybit) at the time of the election. Asset classification was a two-stage process. First, tokens were categorised into the thematic groups DeFi & RWA or AI & Big Data based on their primary utility. Subsequently, a "Made in the U.S." classification was applied if a project's official documentation or website identified its core developers or governing entity as being based in the United States. Finally, the WLF represents the portfolio of the World Liberty Financial.

^a Link to the official webpage of World Liberty Financial: < <https://worldlibertyfinancial.com/>>.

^b Link to website: (<https://intel.arkm.com/explorer/entity/worldlibertyfi>).

Table A2

Results based on the intra-day event window with respect to the event time.

	Overall			Made in U.S.			WLF Portfolio			DeFi & RWA			AI & Big Data		
	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.
Event windows before the event (excluding the event day)															
CAR [-1;event]	1.68 %	3.85	***	1.91 %	2.62	***	3.50 %	2.04	**	0.89 %	1.57		1.49 %	1.28	
CAR [event;0]	0.41 %	1.36		0.35 %	0.74		0.41 %	0.38		0.26 %	0.53		1.18 %	1.34	
CAR [-24h event;event]	2.95 %	5.59	***	3.43 %	3.41	***	4.29 %	2.06	**	1.67 %	2.65	***	3.45 %	2.78	***
CAR [event;+24h event]	1.47 %	3.31	***	1.27 %	1.87	*	1.15 %	0.58		1.26 %	2.03	**	3.01 %	2.21	**
CAR [-1h event;event]	0.63 %	8.79	***	0.66 %	5.78	***	0.53 %	2.04	**	0.58 %	5.21	***	0.71 %	3.41	***
CAR [event;+1h event]	-0.62 %	- 9.96	***	-0.55 %	- 6.12	***	-0.73 %	- 3.35	***	-0.59 %	- 5.68	***	-0.66 %	- 3.79	***
CAR [-24h event;+24h event]	4.72 %	5.65	***	4.95 %	3.12	***	6.47 %	1.62		2.85 %	2.73	***	7.35 %	3.64	***
CAR [-1h event;+1h event]	-0.04 %	- 0.39		0.06 %	0.31		-0.18 %	- 0.59		-0.11 %	- 0.77		0.09 %	0.36	

Notes: *** denotes a statistical significance at a p-value < 0.01, while ** and * denote a statistical significance at a p-value of respectively <0.05 and <0.10. The daily ARs have been winsorized at the 5th and 95th percentiles. The event is identified at the time 05:34 ET (10:34 UTC) on Nov 6 (source: Associated Press).

Table A3

Results based on a sample winsorized at the 1st and 99th percentiles.

	Overall			Made in U.S.			WLF Portfolio			DeFi & RWA			AI & Big Data		
	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.
Event windows before the event (excluding the event day)															
CAR [−10;−1]	−4.81 %	- 6.74	***	−4.94 %	- 3.31	***	−2.12 %	- 1.37		−5.08 %	- 5.82	***	−4.08 %	- 1.80	*
Pre- and post-event window															
CAR [−1;1]	7.76 %	9.82	***	5.74 %	7.10	***	10.63 %	4.58	***	9.69 %	6.81	***	6.26 %	3.63	***
CAR [−1;0]	16.36 %	22.63	***	14.44 %	14.42	***	18.74 %	8.33	***	17.75 %	15.41	***	16.93 %	10.57	***
Event windows after to the event (including the event day)															
CAR [0;2]	1.96 %	2.44	**	0.42 %	0.31		4.04 %	1.51		4.58 %	3.99	***	−2.08 %	- 1.25	
CAR [0;6]	−1.25 %	- 0.94	***	3.89 %	1.25		1.05 %	0.94		−5.23 %	- 4.19	***	−5.40 %	- 1.93	*
CAR [0;30]	37.92 %	8.90	***	50.54 %	5.60	***	25.48 %	3.05	***	37.12 %	5.42	***	28.75 %	4.56	***
Event windows after to the event (excluding the event day)															
CAR [1;2]	−8.06 %	- 11.33	***	−7.47 %	- 5.15	***	−8.08 %	- 3.33	***	−7.17 %	- 6.93	***	−11.39 %	- 9.05	***
CAR [1;6]	−11.28 %	- 7.39	***	−4.00 %	- 1.22		−11.07 %	- 6.68	***	−16.98 %	- 10.48	***	−14.71 %	- 4.88	***
CAR [1;30]	27.89 %	6.20	***	42.65 %	4.49	***	13.36 %	1.76	*	25.37 %	3.51	***	19.44 %	- 8.50	***

Notes: *** denotes a statistical significance at a p-value < 0.01, while ** and * denote a statistical significance at a p-value of respectively <0.05 and <0.10. The daily ARs have been winsorized at the 1st and 99th percentiles.

Table A4

Results based on non-winsorized sample.

	Overall			Made in U.S.			WLF Portfolio			DeFi & RWA			AI & Big Data		
	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.
Event windows before the event (excluding the event day)															
CAR [−10;−1]	−4.81 %	- 6.74	***	−4.94 %	- 3.31	***	−2.12 %	- 1.37		−5.08 %	- 5.82	***	−4.08 %	- 1.80	*
Pre- and post-event window															
CAR [−1;1]	7.99 %	9.21	***	5.74 %	7.10	***	11.65 %	4.03	***	10.18 %	6.28	***	6.23 %	3.61	***
CAR [−1;0]	16.70 %	20.51	***	14.44 %	14.42	***	19.75 %	6.78	***	18.51 %	13.01	***	16.93 %	10.57	***
Event windows after to the event (including the event day)															
CAR [0;2]	2.18 %	2.50	**	0.42 %	0.31		5.05 %	1.71	*	5.08 %	3.74	***	−2.11 %	- 1.26	
CAR [0;6]	−2.90 %	- 1.93	***	2.70 %	0.80		0.99 %	0.89		−7.23 %	- 4.77	***	−8.16 %	- 2.46	**
CAR [0;30]	40.34 %	8.03	***	55.24 %	4.82	***	26.14 %	3.09	***	39.10 %	5.09	***	25.99 %	3.82	***
Event windows after to the event (excluding the event day)															
CAR [1;2]	−8.18 %	- 11.13	***	−7.47 %	- 5.15	***	−8.08 %	- 3.33	***	−7.43 %	- 6.59	***	−11.42 %	- 8.96	***
CAR [1;6]	−13.26 %	- 7.61	***	−5.19 %	- 1.44		−12.15 %	- 5.46	***	−19.74 %	- 10.02	***	−17.47 %	- 4.69	***
CAR [1;30]	29.97 %	5.65	***	47.35 %	3.96	***	13.00 %	1.71	*	26.59 %	3.27	***	16.68 %	- 8.50	***

Notes: *** denotes a statistical significance at a p-value < 0.01. while ** and * denote a statistical significance at a p-value of respectively <0.05 and <0.10.

Table A5

Results based on a mean-adjusted return model.

	Overall			Made in U.S.			WLF Portfolio			DeFi & RWA			AI & Big Data		
	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.	CAR	T-test	Sign.
Event windows before the event (excluding the event day)															
CAR [−10;−1]	−2,40 %	- 2,84	***	−2,69 %	- 1,67	*	0,31 %	0,12		−2,41 %	- 1,98	**	−1,98 %	- 0,83	
Pre- and post-event window															
CAR [−1;1]	17,79 %	25,85	***	16,50 %	17,72	***	20,38 %	15,46	***	18,98 %	15,77	***	17,82 %	13,73	***
CAR [−1;0]	15,67 %	29,85	***	14,90 %	17,28	***	17,18 %	13,17	***	16,20 %	21,91	***	16,49 %	13,53	***
Event windows after to the event (including the event day)															
CAR [0;2]	14,57 %	20,65	***	13,52 %	10,46	***	16,34 %	10,20	***	16,70 %	14,74	***	12,21 %	11,56	***
CAR [0;6]	27,69 %	26,20	***	31,84 %	14,96	***	31,34 %	11,13	***	24,72 %	18,58	***	26,14 %	12,06	***
CAR [0;30]	75,23 %	24,76	***	83,39 %	13,42	***	67,44 %	10,60	***	75,27 %	15,61	***	72,77 %	14,20	***
Event windows after to the event (excluding the event day)															
CAR [1;2]	3,35 %	5,28	***	3,42 %	2,77	***	3,85 %	2,26	**	4,55 %	4,24	***	1,23 %	1,26	
CAR [1;6]	16,47 %	14,86	***	21,74 %	10,02	***	18,84 %	7,10	***	12,56 %	9,22	***	15,16 %	6,11	***
CAR [1;30]	64,01 %	20,21	***	73,29 %	11,03	***	54,94 %	9,13	***	63,12 %	12,74	***	61,79 %	- 5,98	***

Notes: *** denotes a statistical significance at a p-value < 0.01, while ** and * denote a statistical significance at a p-value of respectively <0.05 and <0.10. The daily ARs have been winsorized at the 5th and 95th percentiles.

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